

HookShield: Complete Production Deployment Guide

This guide will walk you through deploying HookShield to a live production environment. Follow every step carefully to get your app running securely on the internet.

Step 1: Rent a Virtual Private Server (VPS)

Since HookShield is containerized using Docker, you need a Linux server to run it.

1. Go to [DigitalOcean.com](https://www.digitalocean.com/) (or Linode / AWS).
2. Create an account and click **Create a Droplet**.
3. Choose **Ubuntu 22.04 (LTS)** as your operating system.
4. Choose the **Basic Plan** (a \$5 to \$10/month plan is perfect to start).
5. Choose a datacenter region closest to where your users are.
6. Under Authentication, select **Password** and create a strong password (save this, you will need it).
7. Click **Create Droplet**. Wait 1-2 minutes for it to boot. You will be given an **IP Address** (e.g., `142.25.12.99`).

Step 2: Get a Domain Name (Optional but Recommended)

To allow users to visit `yourwebsite.com` instead of an IP address:

1. Go to [Namecheap.com](https://www.namecheap.com/) or GoDaddy and buy a domain name.
2. Go to your domain's **DNS Management** page.
3. Add a new **"A Record"**:
 - **Host/Name:** `@` - **Value/Target:** Paste the **IP Address** of your DigitalOcean droplet from Step 1. - **TTL:** Automatic or 30 minutes.

Step 3: Generate Your Real API Keys

Your application needs credentials to send emails and authenticate users securely.

3A. Gmail App Password (For Password Resets)

You cannot use your standard Gmail password. You must generate an App Password.

1. Go to your Google Account Settings -> **Security**.

2. Make sure **2-Step Verification** is turned ON.
3. Search for **App Passwords** in the settings search bar.
4. Create a new App Password named "HookShield".
5. Google will give you a 16-character password (e.g., `abcd efgh ijkl mnop`). Save this.

3B. Google OAuth Client ID (For Google Login)

1. Go to the [Google Cloud Console](https://console.cloud.google.com/).
2. Click "Select a Project" -> **New Project** (Name it "HookShield").
3. Go to **APIs & Services** -> **Credentials**.
4. Click **Create Credentials** -> **OAuth Client ID**.
5. Set Application Type to **Web Application**.
6. Under "Authorized redirect URIs", add: `http://:8000/api/oauth/callback/google` (replace with your domain if you bought one).
7. Copy the **Client ID** and **Client Secret**.

3C. GitHub OAuth Client ID (For GitHub Login)

1. Go to GitHub.com -> Settings -> **Developer Settings** (at the very bottom).
2. Click **OAuth Apps** -> **New OAuth App**.
3. Set the "Homepage URL" to your droplet's IP address or domain.
4. Set the "Authorization callback URL" to: `http://:8000/api/oauth/callback/github`.
5. Click **Register Application**. Copy the **Client ID** and generate a **Client Secret**.

Step 4: Prepare the Server

Now, you will log into your server and install Docker.

1. Open a terminal (Command Prompt or PowerShell) on your personal computer.
2. Type: `ssh root@` and hit Enter.
3. Type `yes` when prompted, and enter the Droplet password you created in Step 1.
4. Run the following command to install Docker and Docker-Compose:

```
``bash apt update && apt install docker.io docker-compose -y ``
```

Step 5: Upload Your Code and Deploy

1. Upload your code to a private GitHub repository.

2. On your VPS terminal, clone the repository:

```
```bash git clone https://github.com/your-username/hookshield.git cd hookshield ```
```

3. Copy the `.env.example` file to create your actual production environment file:

```
```bash cp .env.example .env.production ```
```

4. Open the file to edit it:

```
```bash nano .env.production ```
```

5. Fill in the file with your actual values:

- `JWT_SECRET`: Enter a random long string of text. - `SMTP_USER`: Your Gmail address. - `SMTP_PASSWORD`: The 16-character App Password from Step 3A. - `GOOGLE_CLIENT_ID` / `GITHUB_CLIENT_ID`: The IDs you got in Steps 3B and 3C.

6. Press `Ctrl + X`, then `Y`, then `Enter` to save and exit.

7. Finally, start the application!

```
```bash docker-compose --env-file .env.production up -d --build ```
```

****Congratulations!**** HookShield is now running on your server. Visit your Droplet's IP address (or your domain name) in your browser to see your live application.